

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: Task 2

Student Name: J.Srikar

Roll No.: A24126552083

Date: 25-08-2026

1. TITLE

Design a Styled Webpage Demonstrating HTML5 Canvas and CSS3 Styling with Hover Transitions

2. AIM

To design a web page that demonstrates HTML5 Canvas drawing (rectangles, circles, lines, text) and advanced CSS3 styling features such as linear gradients, flexbox cards, hover transitions, and shadows.

3. OBJECTIVE

- To construct visual divisions of headings and sections using CSS3 style configurations.
- To use gradients (linear-gradients) for backgrounds and shadows (box-shadow) for visual depth.
- To design responsive info-cards styled with dynamic hover transitions.
- To draw basic geometric shapes (rectangles, circles, lines, and text) on HTML5 Canvas using JS 2D contexts.

4. THEORY

CSS3 introduces properties like linear-gradient for smooth color blending, border-radius for rounded corners, and box-shadow for soft shadow offsets. Flexbox provides an aligned container grid system to layout cards dynamically. CSS transitions permit styling changes (e.g. scale scaling, color shifts) to animate smoothly over specified duration values.

The HTML5 <canvas> element uses JavaScript script functions to paint shapes in a browser window. The getContext('2d') method retrieves the 2D workspace to call methods like fillRect(), arc(), lineTo(), and fillText() using pixel coordinate locations.

5. ALGORITHM / PROCEDURE

1. Create a main index file named canvas_styling.html.
2. In style blocks, specify box-sizing, font-family, and define a linear gradient background for the body.

3. Style the header and footer with slate background, white text, and a strong text shadow.
4. Create navigation links, adding background-color transitions on mouse hovers.
5. Construct a centered Canvas container styled with box shadows.
6. Create flex-row card configurations, adding scale transforms (scale(1.05)) on hover.
7. Create buttons inside card columns with background transitions.
8. Define a 600x300 canvas container tag in HTML.
9. Write a JS script block targeting the canvas ID to access its 2D drawing context.
10. Draw a red rectangle, blue circle, green diagonal line, and purple name text using coordinates.
11. Save and verify page load and hover behaviors in a web browser.

6. PROGRAM

canvas_styling.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
  <style>
    * { margin:0; padding:0; box-sizing:border-box; }
    body { font-family:Arial, sans-serif; background:linear-gradient(to
right,#74ebd5,#ACB6E5); padding: 20px; }
    header { background:#2c3e50; color:white; text-align:center; padding:25px; text-
shadow:2px 2px 5px black; border-radius: 5px; }
    header h1 { font-size:32px; font-family:Georgia, serif; }
    nav { background:#34495e; text-align:center; padding:12px; margin: 10px 0; border-
radius: 5px; }
    nav a { color:white; text-decoration:none; margin:15px; font-size:18px; font-
weight:bold; padding:8px 15px; border-radius:5px; transition:0.3s; }
    nav a:hover { background:orange; color:black; }
    .canvas-section { text-align:center; margin:20px; }
    canvas { border:3px solid black; background:white; border-radius:10px; box-
shadow:5px 5px 10px gray; }
    .cards { display:flex; justify-content:center; gap:20px; margin:20px; flex-
wrap:wrap; }
    .card { width:240px; background:white; border:2px solid gray; border-radius:15px;
padding:15px; box-shadow:5px 5px 10px gray; transition:0.4s; }
    .card:hover { transform:scale(1.05); }
    .card h2 { color:darkblue; margin-bottom:10px; font-family:Georgia; font-size:
20px; }
    .card p { margin-bottom:15px; font-size:14px; line-height:1.5; }
    button { background:#3498db; color:white; border:none; padding:8px 16px; border-
radius:8px; cursor:pointer; font-size:14px; transition:0.3s; }
    button:hover { background:#2ecc71; color:black; }
  </style>
</head>
<body>
  <header>
```

```

    <h1>My Styled Webpage</h1>
    <p>HTML5 Canvas and CSS3 Styling Demo</p>
</header>
<nav>
    <a href="#">Home</a>
    <a href="#">About</a>
    <a href="#">Services</a>
</nav>
<section class="canvas-section">
    <h2>HTML5 Canvas</h2>
    <br>
    <canvas id="myCanvas" width="600" height="300"></canvas>
</section>
<section class="cards">
    <div class="card">
        <h2>Card One</h2>
        <p>CSS properties like border, padding, border-radius, and box-shadow.</p>
        <button>Read More</button>
    </div>
    <div class="card">
        <h2>Card Two</h2>
        <p>Hover transformations and transition effects.</p>
        <button>Explore</button>
    </div>
</section>
<script>
    var canvas = document.getElementById("myCanvas");
    var ctx = canvas.getContext("2d");
    ctx.fillStyle = "red";
    ctx.fillRect(40,40,120,70);
    ctx.beginPath();
    ctx.arc(280,75,40,0,2*Math.PI);
    ctx.fillStyle = "blue";
    ctx.fill();
    ctx.beginPath();
    ctx.moveTo(400,40);
    ctx.lineTo(550,120);
    ctx.strokeStyle="green";
    ctx.lineWidth=4;
    ctx.stroke();
    ctx.font="28px Arial";
    ctx.fillStyle="purple";
    ctx.fillText("Srikar",200,220);
</script>
</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<code>linear-gradient(to right, ...)</code>	Defines a horizontal gradient background shifting from light cyan to light purple.
<code>box-shadow: 5px 5px 10px gray</code>	Creates a grey shadow offset behind cards and canvas elements for 3D depth.
<code>transform: scale(1.05)</code>	Scales the target element up by 5% when hovered.
<code>transition: 0.4s</code>	Applies smooth transformations over a 0.4-second interval.
<code>canvas.getContext('2d')</code>	Returns a drawing context on the canvas for rendering 2D shapes.
<code>ctx.fillRect(40,40,120,70)</code>	Draws a filled red rectangle at coordinates (40, 40) with width=120, height=70.
<code>ctx.arc(280,75,40,0,2*Math.PI)</code>	Paints a circle centered at (280, 75) with radius=40 using arc paths.
<code>ctx.lineTo(550,120)</code>	Draws a line segment from (400, 40) to (550, 120).
<code>ctx.stroke()</code>	Applies color to the drawn line path using stroke styles.
<code>ctx.fillText('Srikar', 200, 220)</code>	Renders name text 'Srikar' inside the canvas at coords (200, 220).

8. TEST CASES AND EXECUTION DATA

Test Case	Feature Tested	Action / Input	Expected Result	Actual Result	Status
TC-01	Page Load	Load canvas_styling.html	Gradient background, header, navigation links, and cards render	Rendered successfully	Pass
TC-02	Canvas Drawings	Check canvas content	Red rect, blue circle, green line, and name text are drawn	Drawn correctly	Pass

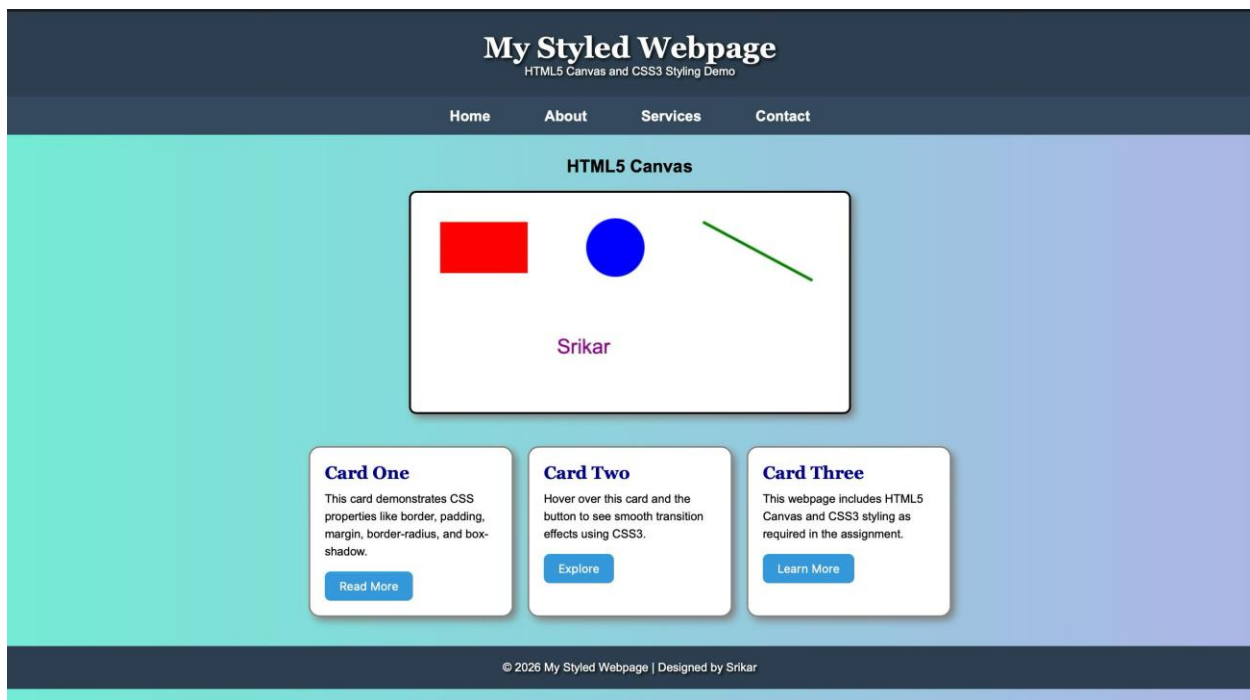
TC-03	Card Hover	Move mouse cursor over a card	Card scales up by 5% smoothly with increased shadow	Card scaled smoothly	Pass
TC-04	Button Hover	Move mouse cursor over a button	Button background turns green and text turns black	Color changed successfully	Pass

Additional Validation Test

Test Case	Input	Expected Result	Status
TC-05	Hover Nav Links	Link background turns orange smoothly on hover	Pass

9. OUTPUT

Output : HTML5 Canvas and CSS3 Webpage Rendering



10. RESULT

Thus, a styled HTML5/CSS3 webpage containing dynamic linear-gradient backdrops, hover-animated cards, and coordinate drawings on Canvas (rect, circle, line, name text) was successfully created and verified.